

Syed Muhammad Ayyan Ibrar

AI/ML Engineer | Python, Data Pipelines & Backend Systems

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SUMMARY

AI/ML Engineer with hands-on experience building Python-based data pipelines, backend systems, and end-to-end ML workflows. Experienced in data ingestion, processing, validation, database integration, and production deployment. Strong Python and PostgreSQL foundation with experience building systems that transform and retrieve structured and unstructured data.

RELEVANT SKILLS

Programming & Backend: Python, FastAPI, Pydantic, Git, Docker

Data Engineering: Data Ingestion, Data Cleaning, Data Validation, Data Transformation, ETL Pipelines, Unstructured Data Processing

Databases: SQL, PostgreSQL, ChromaDB, Qdrant, Redis

Cloud: Azure, GCP

AI/ML: Machine Learning, RAG, Embeddings, LLM Applications, Model Evaluation

EXPERIENCE

Freelance AI/ML Engineer - Self-Employed *2024 – Present*

- Built end-to-end Python data and ML pipelines covering data ingestion, preprocessing, validation, model training, evaluation, and production deployment.
- Designed data-processing workflows to transform and prepare heterogeneous datasets for downstream machine learning and AI systems.
- Developed backend services and automation workflows integrating data processing, databases, and model inference into production systems.

PROJECTS

NEXUS - Autonomous Multi-Agent Research Pipeline - github.com/syedmuhammadayyanibrar/nexus-intelligence-os

- Built a Python pipeline that ingests information from live web sources, processes extracted content through multiple stages, and stores information in ChromaDB for downstream retrieval.
- Orchestrated a 6-agent LangGraph workflow for planning, extraction, contradiction detection, code execution, synthesis, and validation.
- Integrated FastAPI Server-Sent Events with a Streamlit interface to stream pipeline execution and agent progress in real time.

Decentralized Multi-Agent Negotiation System - github.com/syedmuhammadayyanibrar/negotiation_agent

- Engineered autonomous LLM agents using Groq LLaMA 70B, game-theoretic negotiation protocols, adaptive trust modeling, and coalition formation in a fully decentralized environment.
- Developed an evaluation harness comparing agent strategies using Jain fairness, the Gini coefficient, and Nash bargaining distance across repeated negotiations.

RedLLM - Automated Red-Teaming Scanner for LLM Applications - github.com/syedmuhammadayyanibrar/RedLLM

- Built a CLI security evaluation framework testing LLM, RAG, and agent applications across 6 OWASP-mapped attack categories, with a CVSS-inspired 1-10 severity scoring system validated against matched vulnerable/hardened test cases.

Metacognitive Agent - Confidence-Aware Query Router - github.com/syedmuhammadayyanibrar/metacognitive-agent-v2

- Built a confidence-aware LLM query router combining verbalized confidence, consistency, and retrieval signals to route requests between answering, retrieval, deferral, and rejection.
- Implemented Expected Calibration Error evaluation to measure confidence reliability and identify poorly calibrated model behavior.

EDUCATION

BS, Artificial Intelligence - The Islamia University of Bahawalpur

2022 – 2026 | CGPA: 3.76

Advanced AI Bootcamp - GIKI Institute (GIKI), Topi, Swabi

Grade A | LLM Engineering, RAG, Agentic AI, MLOps